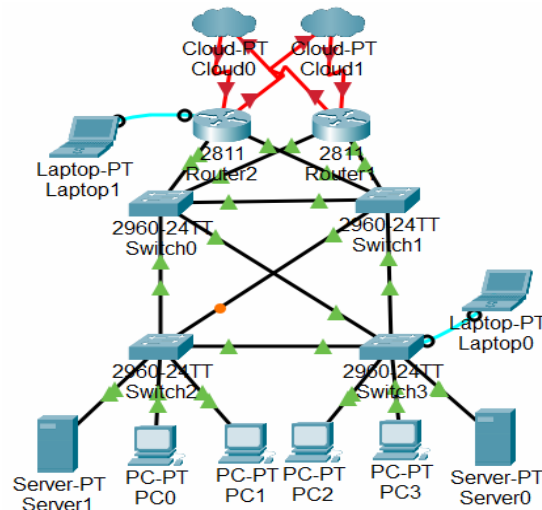


Project description

You are a system administrator for a large company. You were get a task from your employer to set up a redundant LAN network to forward the network traffic of your company. When operating the network, special care must be taken for secure operation, so you must configure the switches and routers such a way, so that provide the appropriate redundancy between the VLANs and in the network communication, based on the parameters described below!

(Before commissioning (for testing), you must perform the network simulation in Packet Tracer, so that you would have to solve the problems of device configuration or testing of proper operation not on the real live network!)



1. You must set the hostname of the switches to Birthplace_CentAS1 – 4 and the hostname of the routers to Birthplace_CentR1 – 2 (for example, if you were born in Pécs, then Pecs_CentAS1 – 4 and Pecs_CentR1 – 2)!
2. Set the enable password and password encryption on all switches and routers. (The password for the switches should be CentSwitch1 – 4 and the password for the router should be CentRouter1 – 2!)
3. Since secure operation is a priority, set SSH on all switches and routers. (The domain, in which the devices are located is company.hu.) To increase security, use SSH version 2!
4. To increase security, user-based authentication is required, so create a user with an Admin and a NormalUser ID with the appropriate rights and settings, on all devices! (Passwords: Admin1 – 4: Administrator1 – 4, NormalUser1 – 4: user1 – 4)
5. Set the accesses! Use user-based authentication for both the terminal and the console access! Only accept the SSH-based communication on the terminal connections!
6. For safe operation, you must create the following VLAN structure on all switches!
 - VLAN 1X: IT (a), Operation (b),
 - VLAN 2X: Staff (a), HR (b),
 - VLAN 3X: Production (a), Manufacturing (b),
 - VLAN 4X: EO (a), EcoOff (b),
 - VLAN 9X: Management (a), MGMNT (b),
 - VLAN 10X: Other (a), Spare (b), where X: the day of your birthday date, rounded to the nearest 10: (Eg: If you were born on December 13th, 13 is the day, and according to the rounding rule, the given value is 10, which means that 3X is 30. If you were born on 18th, then this value is 20, due to rounding rule, so 3X is 60 in this case.)
7. Configure the VTP protocol on the switches! Configure the first two switches as a VTP server (a) and the second two as a client (a)! (In the case (b) reverse!) (The name of the VTP domain should be company, the password should be: companyvtp!)

8. For proper operation, divide the FastEthernet ports of the switch between VLANs as follows: (In each VLAN, end-devices are connected to the ports!)
 - VLAN 1X: 1 – 3 port (a), 24 – 21 (b),
 - VLAN 2X: 4 – 7 port (a), 20 – 17 (b),
 - VLAN 3X: 8 – 11 port (a), 16 – 13 (b),
 - VLAN 4X: 12 – 15 port (a), 12 – 9 (b).to increase the security, move unused ports to Other (a) or Spare (b) VLANs and shutdown them.
enable bpduguard service on access ports!
Additionally, set the port security on the ports you are using, with the following parameters:
 - Sticky MAC address management,
 - maximum 3 address.
9. Configure the spanning-tree protocol on the switches with the following parameters:
 - Set the portfast service on the access ports on all switches!
 - Set the first switch as the primary root switch for 1X, 2X, and 3X VLANs, and the secondary root switch for 4X and 9X VLANs. In (b) use the third switch instead of the first!
 - Set the second switch as the secondary root switch of the 1X, 2X, and 3X VLANs, and to the primary root switch of the 4X and 9X VLANs. In (b) use the fourth switch instead of the second!
10. Set the SVI for the Management (a) (MGMNT (b)) VLAN on each switch, to the IP address from 192.168.9X.247 to 192.168.9X.250!
11. Set the default gateway address of the switches to 192.168.9X.254!
12. Connect the all switches with each other, to form a redundant switch network! Use the last FastEthernet (a) and the first FastEthernet (b) and GigabitEthernet ports for trunk connections! Configure the interfaces for a proper trunk connection! (Only allow VLANs already created on trunk interfaces to be forwarded, and change the native VLAN to the Management (a) (MGMNT (b)) VLAN!)
13. Configure the routing between VLANs, on routers! (Use the Router-on-a-Stick method! The ID of the sub-interfaces must match the ID of the VLAN! The IP addresses must be 192.168.VLAN_ID.252 on CentR1 and 192.168.VLAN_ID.253 on CentR2, with the appropriate subnet mask!)
14. Configure the HSRP with load-sharing on routers, with the following parameters:
 - CentR1 should be the active router in 1X, 2X and 3X VLANs, and standby in 4X and 9X VLANs!
 - CentR2 should be the active router in 4X and 9X VLANs, and standby in 1X, 2X, and 3X VLANs!
 - The IP address of active routers should be 192.168.VLAN_ID.254, and the IP address of standby routers should be 192.168.VLAN_ID.1!
 - Standby routers must be able to take over the active role if necessary, and active routers must be able to take back this, if they are working properly!
 - In the case (b) the roles of the two routers are reversed!
15. Save the configuration file of all devices, so that these files are not lost when the devices are restarted!
16. Test the proper operation of the settings!
17. Prepare the proper documentation of the task!